

Agile Product Portfolio
Software Engineering Group Coursework

Student Help - A University Student Support Platform

Group number: 4

Student ID	Student Name
3059787	Yiran Wang
2841331	Tao Xie
2938099	Yi Wu
3020084	Daoming Han

1. Planning (20%)

Conceptual model

Personas → Product Vision → User Stories → Features → Prototype → Tests

1.1 Personas

Persona Name	Role/User type	Goals	Pain Points	Related User Stories (IDs)
Amira Khan	Student	Receive messages, complete surveys, book appointments.	Important notices are spread across email and chat; students are unable to provide feedback; booking help is confusing	S1、 S2、 S7、 S9
Dr. Lewis	Teacher	Publish urgent and non-urgent messages, correct or delete outdated or error messages, manage appointment slots efficiently	Students miss key notices; resending updated messages is repetitive; manual booking coordination service support is time-consuming.	S3、 S4、 S5、 S8、 S10
Ms. Chen	Administrator	Monitor survey completion and results, identify satisfaction issues and support service quality improvement.	Survey data arrives in separate files; difficult to gain a quick overview of student feedback trends.	S6

Note: You need to have at least two personas. The personas should be used to justify the product vision and user stories.

1.2 Product Vision

FOR (target customer) Students who need timely support, teachers who provide guidance, and administrators who monitor student feedback

WHO (statement of the need or opportunity) Need a single platform to communicate support information, collect questionnaire responses and coordinate appointments efficiently

The **(PRODUCT NAME)** is a (product category) Student Support Platform is a student academics, well-being, and careers

THAT (key benefit, compelling reason to buy) Seamlessly integrates targeted messaging with AI summaries, multi-dimensional surveys, and an automated appointment booking system for academics, mental health, and career counseling.

UNLIKE (primary competitive alternative) Fragmented legacy systems, manual scheduling tools, or using separate third-party apps for emails, forms, and bookings.

OUR PRODUCT (statement of primary differentiation) Combines information delivery, AI-generated message summaries, structured questionnaires and appointment workflow management in one coherent service.

Note: This product vision should be derived from the defined personas and guides backlog prioritisation and design decisions.

1.3 Meeting Logs (Appendix)

2. Story Writing (20%)

2.1 Product Backlog and User Stories

Story ID	Feature (epics)	Story Title	User Story	Acceptance Criteria	Priority (Must / Should / Could)	Justification of prioritisation /Trade-off	Story Owner
S1	Message Management	Publish messages	As a teacher, I want to publish a message that includes the subject, content, urgency level, and the option to upload a file so that students can receive the information promptly.	Given a teacher is logged in, When a valid message form is submitted, then the message is saved with teacher details and any uploaded files are linked to it.	Must	Core behavior	Tao Xie
S2	Message Management	Edit and delete messages	As a teacher, I want to edit and republish or delete messages so that outdated or error information can be corrected quickly.	Given the current user owns the message, When they edit/delete it, then the content/files are updated or removed and unauthorised users are blocked.	Must	Improve the accuracy and quality of message	Tao Xie
S3	Message Management	Filter and search messages	As a student, I want to be able to view, filter, and search information so that I can quickly find urgent or non-urgent and important information.	Given published messages, When a student filter search, then matching results are shown.	Must	Core behavior	Tao Xie

S4	Message Management	Summarize message with AI assistant	As a student, I want AI to summarize messages so that I can efficiently grasp key points and save time.	Given published messages, When a student click AI assistant, then a short AI summary are shown.	Must	Improves message reasability	Yiran Wang
S5	Survey	Complete questionnaires	As a student, I want to fill in study progress, school management satisfaction and teaching satisfaction questionnaires so that my feedback can be recorded.	Given a student selects a questionnaire, When all required questions are completed and submitted, then the response is stored successfully.	Must	Boosts feedback clarity	Yi Wu
S6	Survey	Review questionnaire results	As an administrator, I want to receive questionnaire results so that I can monitor student concerns and support service quality.	Given student survey submissions exist in the system, When the admin opens a results page, then averages/counts/comments are displayed for the selected survey category.	Must	Enhances admin oversight	Yi Wu
S7	Appointment management	Publish appointment slots	As a teacher, I want to publish available appointment times so that students can book support without back-and-forth messaging.	Given a teacher selects a date and one or more slots, when the form is submitted, then non-duplicate slots are created and visible on the management page.	Must	The starting point of the booking workflow; without availability, students cannot book.	Daoming Han

S8	Appointment management	Submit booking request	As a student, I want to choose support teacher and time slot and submit a booking so that I can request help in a structured way.	Given available slots exist, When a student selects learning, mental health or employment support and submits the form, Then the appointment is successfully recorded.	Must	Connects students with teachers	Daoming Han
S9	Appointment management	View appointments	As a teacher, I want to view all appointment requests submitted by students, so that I can prepare in advance and manage my schedule effectively.	Given students have submitted bookings, when I open "My Appointments" list, Then I can see the student's name, the appointment category, and the specific time.	Must	Ensures teachers receive the requests and can manage their workload	Daoming Han
S10	Appointment management	Cancel appointment	As a student, I want to cancel an existing appointment, So that I can free up the time slot for other students if my plans change.	Given I have an upcoming booked appointment, When I click "Cancel Appointment" and confirm, Then this appointment will not be displayed on the "My Appointments" page.	Should	Improves resource utilization	Daoming Han

2.2 Sprint 1 Summary

Item	Sprint 1 (Weeks2-6) Discovery, MVP design and partial implementation	Sprint 2(Weeks7-11) Refinement, testing & evaluation
Planned stories (Plan - P)	S1, S3, S5, S7, S8	S2, S4, S6, S9, S10
Delivered stories (Done - D)	S1, S3, S5, S7, S8	S2, S4, S6, S9, S10
Stories not completed (No - N)	N/A	N/A
Key change planned for Sprint 2	Focus on closing feedback and booking loops	N/A

Note: Stories marked “P” or “N” should be reflected upon in Sprint learning or retrospectives.

2.3 Group contribution and Individual contribution (Appendix)

3. Design (20%)

3.1 Design Models and Evidence

Diagram	Type (Class / Sequence)	Purpose	Related User Stories (Story IDs)	Design decision rationale
D1	Class	Domain Structure	S1, S2	Clarifies data structure
D1	Class	Domain Structure (User, Teacher, Student, Administrator, Message, Survey, Appointment)	S1, S2, S3, S4, S5, S6, S7, S8, S9, S10	Defines core system entities and relationships, ensuring all features (messaging, survey, booking) are structurally supported.
D2	Sequence	To show the process of a teacher publishing a message with subject, content, urgency level, and optional file upload.	S1	This sequence was designed to clearly show how a teacher submits a message form, how the system validates the input, stores the message with teacher details, and links any uploaded files. This supports S1 because publishing messages is the core starting point of the message management feature.
D3	Sequence	To show the workflow for editing, republishing, or deleting an existing message.	S2	This sequence focuses on checking message ownership before allowing changes. The design ensures that only the teacher who created the message can edit or delete it, while unauthorised users are blocked. This supports message accuracy and prevents incorrect or unauthorised updates.
D4、 D5	Sequence	To show how students view, filter, and search published messages.	S3	This sequence was designed to show how the system retrieves published messages and returns matching results based on search keywords, urgency level, or importance. It supports S3 because students need an efficient way to find relevant or urgent information instead

				of browsing all messages manually.
D6	Sequence	To show how a student uses the AI assistant to summarise a published message.	S4	This sequence separates the message retrieval process from the AI summarisation process. The design allows the system to first identify the selected message and then send its content to the AI assistant to generate a short summary. This supports S4 by improving message readability and helping students understand key points quickly.
D7	Sequence	To show the process of a student completing and submitting a questionnaire.	S5	This sequence was designed to show how a student selects a questionnaire, completes required questions, submits responses, and receives confirmation that the response has been stored. It supports S5 because student feedback must be recorded accurately before it can be reviewed or analysed later.
D8	Sequence	To show how an administrator reviews questionnaire results.	S6	This sequence shows how the administrator opens the results page, selects a survey category, and receives aggregated results such as averages, counts, and comments. The design separates individual student submissions from admin-level result viewing, which supports privacy, clarity, and effective monitoring of student concerns.
D9	Sequence	To show how a teacher publishes available appointment slots.	S7	This sequence was designed to show how a teacher selects dates and time slots, how the system checks for duplicate slots, and how valid slots become visible on the management page. This supports S7 because appointment availability must exist before students can make any booking requests.

D10	Sequence	To show how a student submits a booking request for a support appointment.	S8	This sequence shows how a student selects a support type, teacher, and available time slot before submitting a booking request. The design connects students with suitable teacher availability and records the appointment in a structured way. This supports S8 because it turns available slots into actual appointment requests.
D11	Sequence	To show how a teacher views appointment requests submitted by students.	S9	This sequence was designed to show how the teacher opens the “My Appointments” list and how the system retrieves booking details such as student name, appointment category, and time. This supports S9 because teachers need a clear overview of requests in order to prepare in advance and manage their workload.
D12	Sequence	To show how a student cancels an existing appointment.	S10	This sequence shows how a student selects an upcoming appointment, confirms cancellation, and how the system removes the appointment from the student’s appointment list. The design supports better resource utilisation because cancelled appointments should no longer appear as active bookings and may allow the time slot to become available again.

Note: You are **not penalised** for modifying or discarding models if justified by learning.

3.2 Stories and Evidence

For each story marked as “Done” in Sprint 1 must be mapped to at least one test.

Story ID	Evidence Type (UML Diagram / Git Commit ID(s) / Test Evidence: testcases / their results)	Reference (file / commit / screenshot)	Acceptance Criteria Met (Yes / Partially / No)	Notes (if partially completed)
S1-S10	Git commit	a691b4f	Yes	

Note: If a story is marked “Done” without evidence, it will not be treated as complete.

4. Working Prototype and Implemented Features (20%)

Submit a **short video (3-5 minutes)** that demonstrates the implementation of 2 core end-to-end features + refinement of Sprint 1 feature. Each demonstrated feature must be traceable to user stories and acceptance criteria. The video should be submitted under the “**Product Video**” assignment on Canvas.

5. Testing and Evaluation (20%)

Story ID	Acceptance Criteria	Test Performed	Outcome (Pass / Partial / Fail)	Accessibility Checked	Notes (If partially tested)
S1	Teacher can publish a new message (with optional attachment)	Automated pytest(test_create_message_success, test_create_message_with_file) + manual test	Pass	Keyboard only	
S1	Published message appears in student's message list	Automated pytest (test_messages_page_load) + manual test	Pass	Keyboard only	
S2	Teacher can edit their own message content	Automated pytest(test_edit_message) + manual test	Pass	Keyboard only	
S2	Teacher can delete their own message	Automated pytest(test_delete_message) + manual test	Pass	Keyboard only	
S2	Teachers cannot edit/delete other teachers' messages	Automated pytest(test_teacher_cannot_edit_others_message) + manual test	Pass	Keyboard only	

S3	Students can filter messages by urgency	Automated pytest(test_messages_page_load) + manual test	Pass	Keyboard only	
S3	Students can search messages by keyword	Automated pytest(test_student_message_search) + manual test	Pass	Keyboard only	
S4	Student can use AI assistant to summarize messages	Automated pytest(test_student_visit_ai, test_ai_summarize_function) + manual test	Pass	Keyboard only	
S4	Student can use AI assistant to extract deadlines	Automated pytest(test_student_can_access_ai, test_student_can_access_ai, test_ai_deadline_reminder) + manual test	Pass	Keyboard only	
S4	Student can use AI assistant to get motivational messages	Automated pytest(test_ai_motivation) + manual test	Pass	Keyboard only	
S5	Student can complete learning, management, and teaching questionnaires	Automated pytest(test_survey_home_student_access, test_post_select_learning_survey) + manual test	Pass	Keyboard only	
S5	Questionnaire responses are saved to the database	Automated pytest(test_survey_response_creation) + manual test	Pass	Keyboard only	

S6	Admin can view all survey results	Automated pytest(test_admin_results_access_control, test_management_survey_results) + manual test	Pass	Keyboard only	
S6	Non-admin users cannot access survey results	Automated pytest(test_admin_results_access_control) + manual test	Pass	Keyboard only	
S7	Teacher can publish available appointment slots	Automated pytest(test_teacher_can_manage_time_slots) + manual test	Pass	Keyboard only	
S7	Teacher's time slots are visible to students	Automated pytest(test_teacher_view_student_appointments) + manual test	Pass	Keyboard only	
S8	Student can submit a booking request for an available slot	Automated pytest (test_student_book_different_subject_teachers, test_different_students_book_same_teacher)+ manual test	Pass	Keyboard only	
S8	Students cannot book already taken or past slots	Automated pytest(test_student_can_view_appointment_page) + manual test	Pass	Keyboard only	
S9	Teacher can view all student appointments	Automated pytest(test_teacher_view_student_appointments) + manual test	Pass	Keyboard only	
S10	Student can cancel their own appointment	Automated pytest(test_student_cancel_appointment) + manual test	Pass	Keyboard only	

S10	Cancelled appointment frees up the teacher's time slot	Automated pytest(test_student_cancel_appointm ent) + manual test	Pass	Keyboard only	
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Note: You may have more than one test per story. At least one test per story is required.

APPENDIX

A. Group Contribution

Story ID	Story Title	Predicted workload in hours	Actual workload in hours	Status (TODO/In progress / DONE)	Owner name
S1	Publish messages	6	4	DONE	Tao Xie
S2	Edit and delete messages	4	3	DONE	Tao Xie
S3	Filter and search messages	8	10	DONE	Tao Xie
S4	Summarize message with AI assistant	5	8	DONE	Yiran Wang
S5	Complete questionnaires	6	8	DONE	Yi Wu
S6	Review questionnaires	8	10	DONE	Yi Wu
S7	Publish appointment slots	6	8	DONE	Daoming Han
S8	Submit booking request	7	8	DONE	Daoming Han
S9	View appointments	4	3	DONE	Daoming Han
S10	Cancel appointments	5	5	DONE	Daoming Han

B. Individual Contribution

Team member	Contribution (%)	Key Contributions	Evidence
Yiran Wang	25%	User Story S4, frontend and backend of AI Assistant, testing of three major modules, Demo recording, and documentation writing	commit 6696942
Xie Tao	25%	Meeting minutes, User Stories S1-S3, front-end and back-end of information modules, document writing	commit e43d21b
Yi Wu	25%	User Story S5-6, Frontend and Backend of Questionnaire Module, UML Diagram	commit f681597
Daoming Han	25%	User Stories S7-S10, Frontend and Backend of the Reservation Module, ProcessScreenshotRecord	commit DA8ddl1

C. Meeting Logs

You are required to schedule at least 2 meetings with your Customer (TA).

Meeting ID	Team attendees	Customer (TA's name)	When	Agenda items	Actions and Owner

M1	Yiran Wang Tao Xie Yi Wu Daoming Han	Aju Ani Justus	19/02/2026 14:40-15:00	1. It presents the initial user stories, user personas, class diagrams and use case diagrams independently completed by each team member. 2. The teaching assistant reviewed the project ideas, approved the project scope, and confirmed the student support system centered on message communication. 3. The team unanimously decided to start the development with the core message module.	- All members: Refine and finalize user stories based on the teaching assistant's feedback - Xie Tao: Set up the Flask project structure and start developing the message module
M2	Yiran Wang Tao Xie Yi Wu Daoming Han	Aju Ani Justus	26/02/2026 15:40-16:00	1. Demonstrate the completed core message functions (creation, viewing, and default list display), with the login function not yet integrated. 2. Introduced the implementation of message filtering and advanced retrieval under multiple filtering conditions (such as message title, priority, sorting method, etc.). 3. Discussed subsequent functions: file upload,	- Tao Xie: Develop the file upload function and independent file storage page - Yi Wu: Research and design the questionnaire (later renamed survey) function - Yiran Wang: Further edit user stories and draw UML diagrams

				questionnaire, and reservation functions. 4. The teaching assistant approved the current message module and agreed to further expand the planned functions.	
M3	Yiran Wang Tao Xie Yi Wu Daoming Han	Aju Ani Justus	24/03/2026 15:40-16:00	1. Demonstrate the implemented login system and complete role separation for teachers and students (students are not allowed to create new messages). 2. Demonstrate the file upload function (upload records are currently stored in JSON files, and the issue of overwriting files with duplicate names has not been resolved). 3. Show the preliminary survey function. 4. Teaching assistants put forward improvement suggestions regarding file security and permission control.	- Tao Xie: Restructure the file upload logic, add timestamp renaming to avoid file overwriting, and improve role permission control across all pages. - Yi Wu: Integrate the survey function with the existing framework. - Yiran Wang: Select models required for the local large model and develop the AI assistant function.
M4	Yiran Wang Tao Xie Yi Wu Daoming Han	Aju Ani Justus	27/03/2026 12:40-13:00	1. Present the complete project documentation, including various charts and development logs.	- Tao Xie, Han Daoming: Resolve Git merge conflicts and integrate team code - All members: Complete final testing and interface optimization - Yiran Wang: Organize all documents,

				2. Demonstrate all core functions: messages, file upload, AI assistant, reservation, and survey. 3. Demonstrate the survey function: after students submit their responses, teachers can view the summarized results on the teacher side. 4. Discuss the final submission requirements, including Git code merging, README files, and other related matters.	README files, submission materials (demo presentation videos) and other related content
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Meeting Diary for Team meetings				
Meeting ID	Team attendees	When	Agenda items	Actions and Owner
M1	Yiran Wang Tao Xie Yi Wu Daoming Han	27/03/2026 9:00-24:00	Determine the project theme: Student Help - University Student Support Platform. Identify primary users: students, teachers/staff, and administrators. Initially define three core functional modules: message, survey, and appointment. Clarify that the system needs to distinguish between teacher and student accounts first.	- Tao Xie: Responsible for the requirement analysis of the Message module, including teachers sending messages and uploading files, and students viewing messages and downloading files. - Yi Wu: Responsible for the requirement analysis of the Survey module, including students filling out questionnaires. - Daoming Han: Responsible for the requirement analysis of the Appointment module, covering three types of reservations: academic study, mental health counseling, and career guidance. - Yiran Wang: Responsible for registration, login and user role design, including student, teacher and administrator.
M2	Yiran Wang Tao Xie Yi Wu Daoming Han	03/04/2026 9:00-24:00	Refine system functions and user stories. Discuss the permissions of teachers and	- Xie Tao: Design the Message interface, including subject, content, priority, publish time and upload files;

			students in different modules. Add edit, delete, priority, filter and search functions to the Message module. Expand the Survey module from one questionnaire to three questionnaires. Add appointment type, appointment time and problem description to the Appointment module, and classify appointments by different teachers.	teachers can only view the messages they have released. - Yi Wu: Design three entrances for Survey: Learning Status Survey, School Management Satisfaction Survey, and Teaching Satisfaction Survey for Teachers. - Daoming Han: Design the Appointment form, covering appointment type, calendar time and problem description; classify by Lecturer, Wellbeing Adviser and Careers Adviser. - Yiran Wang: Design the registration and login interface, including username, email, school ID, password, confirm password, role and teacher type.
M3	Yiran Wang Tao Xie Yi Wu Daoming Han	10/04/2026 9:00-24:00	Discuss the optimization during the development process. The Message module integrates an AI assistant to summarize current messages, which can automatically generate summaries after setting a message quantity threshold. In the Survey module, students can view questionnaire results after submitting their responses. In the Appointment module, teachers can check students' reservation submissions after students complete their appointments.	- Tao Xie: Develop the functions of publishing, editing, deleting, read/unread status, priority sorting and searching for Message. - Yi Wu: Develop the function for teachers to view reservation results in Survey. - Daoming Han: Develop the functions of submitting reservations and receiving reservation notifications for Appointment. - Yiran Wang: Develop AI Assistant, permission control and database connection, ensuring that different roles can only access their corresponding content.
M4	Yiran Wang Tao Xie Yi Wu Daoming Han	17/04/2026 9:00-24:00	Complete system integration and testing. Verify that the student side can receive messages, submit	- Tao Xie: Test the Message module, including teachers releasing messages, students viewing messages, file upload and download, and AI summary.

			surveys, and create appointments. Check that the teacher side can publish and manage their own messages, view survey outcomes, and receive appointment results. Prepare the final presentation to illustrate how the system supports students' academic learning, mental health, and career counseling services.	- Yi Wu: Test the Survey module, including three types of questionnaires, student submission, and teachers viewing results. - Daoming Han: Test the Appointment module, including three types of reservations: study, mental health, and career network. - Yiran Wang: Complete the overall UI inspection, role login test, administrator permission test, and organize the final demo.
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Note: Meeting logs are not marked directly and are used only as supporting evidence.